

Assignment I

(Multiple Integrals)

Subject: Calculus (MTH 210)

Deadline: 2080-08-24

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1. Evaluate:

(i) $\int_0^1 \int_0^2 xy \, dydx.$

Ans:

(ii) $\int_0^\pi \int_0^x x \sin y \, dydx.$

Ans: $\frac{4 + \pi^2}{2}$

2. Sketch the region of integration and evaluate the following integrals by changing the order of integration.

(i) $\int_0^2 \int_x^2 \sin xy \, dydx$

(ii) $\int_0^{a/\sqrt{2}} \int_y^{\sqrt{a^2-y^2}} x \, dx dy$

Ans: $\frac{a^3\sqrt{2}}{6}$

(iii) $\int_0^{\frac{a}{\sqrt{2}}} \int_y^{\sqrt{a^2-y^2}} \log(x^2 + y^2) \, dx dy, (a > 0)$

Ans: $\frac{\pi a^2}{4}(\log a - \frac{1}{2})$

3. Using the double integration find the area of the region bounded by the curves $y = \sin x$, the line $x = 0$ and $x = \frac{\pi}{2}$. **Ans:** 1 sq. unit

4. Find the area bounded by the parabolas $x = y^2 - 1$ and $x = 2y^2 - 2$ by using double integration.

5. Find the volume in the first octant bounded by co-ordinates planes, the cylinder $x^2 + y^2 = 4$ and plane $z + y = 3$. **Ans:** $3\pi - \frac{8}{3}$

6. Find the volume of the tetrahedron bounded by the co-ordinate planes and the planes $x + y + z = 1$.

7. Find the volume of the cylinder $x^2 + y^2 - 2ax = 0$ intercepted between the parabolic $x^2 + y^2 = 2az$ and the xy - plane. **Ans:** $\frac{3}{4}\pi a^3$

8. Find the volume of the solid cut from the first octant by the surface $z = 4 - x^2 - y$. **Ans:** $\frac{128}{15}$

9. Find the volume of the solid in the first octant bounded by the co-ordinate planes, the planes $x = 3$, and the parabolic cylinder $z = 4 - y^2$. **Ans:** 16

10. Evaluate by changing into polar integral.

$$(i) \int_0^3 \int_0^{x\sqrt{2}} \frac{dydx}{\sqrt{x^2+y^2}}. \quad \text{Ans: } 3[\log(2+\sqrt{3})]$$

$$(ii) \int_0^{\frac{a}{\sqrt{2}}} \int_y^{\sqrt{a^2-y^2}} \log(x^2+y^2) dx dy, (a > 0). \quad \text{Ans: } \frac{\pi a^2}{4}(\log a - \frac{1}{2})$$

$$(iii) \int_0^2 \int_0^{\sqrt{4-x^2}} \frac{xy}{\sqrt{x^2+y^2}} dy dx. \quad \text{Ans: } \frac{4}{3}$$

$$(iv) \int_0^{4a} \int_{y^2/4a}^y \frac{x^2-y^2}{x^2+y^2} dx dy. \quad \text{Ans: } 8a^2(\frac{\pi}{2} - \frac{5}{3})$$

11. Define the double integral of a function of two variable over a bounded region in the co-ordinate region. Does the order of integration matter? How are the limit of integration determined? Give examples.

12. Why we change Cartesian integral to polar integral? How would you evaluate the double integral of a continuous function $f(x, y)$ over the region R in the xy -plane enclosed by the triangle with vertices $(0, 1)$, $(2, 0)$, and $(1, 2)$? Give reason for your answer.

13. Evaluate the following triple integral:

$$(i) \int_0^1 \int_0^z \int_{x-z}^{x+z} (x+y+z) dy dx dz. \quad \text{Ans: } 1$$

$$(ii) \int_1^e \int_1^{\log y} \int_1^{e^x} \log z dz dx dy. \quad \text{Ans: } \frac{1}{4}(e^2 - 8e + 13)$$

$$(iii) \int_0^{\pi/2} \int_0^{a \sin \theta} \int_0^{\frac{a^2-r^2}{a}} r dz dr d\theta. \quad \text{Ans: } \frac{5\pi a^3}{64}$$

14. Evaluate $\iiint x^{l-1} y^{m-1} z^{n-1} dx dy dz$, where x, y, z are all positive but $(\frac{x}{a})^p + (\frac{y}{b})^q + (\frac{z}{c})^r \leq 1$.

$$\text{Ans: } \frac{a^n b^m c^n}{pqr} \frac{\Gamma \frac{l}{p} \Gamma \frac{m}{q} \Gamma \frac{n}{r}}{\Gamma(\frac{l}{p} + \frac{m}{q} + \frac{n}{r} + 1)}$$

15. Find by triple integration, volume of sphere $x^2 + y^2 + z^2 = a^2$. Ans: $\frac{4\pi a^3}{3}$